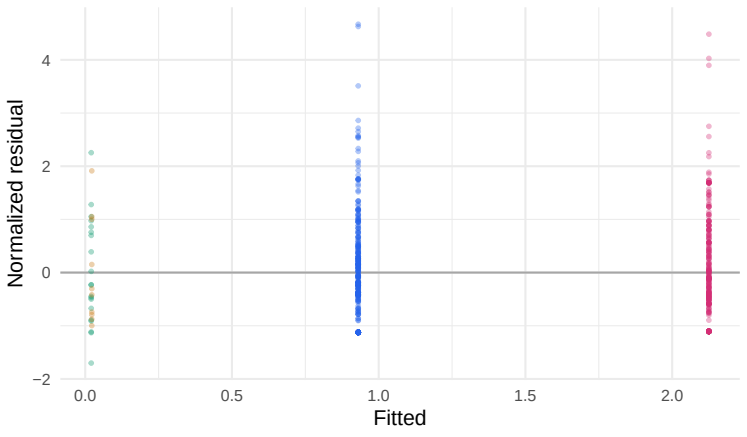
MA(1) offset−model residual checks: Observed community turnover

One row per offset: scatter / Q−Q / Holocene lag products for the selected MA(1)

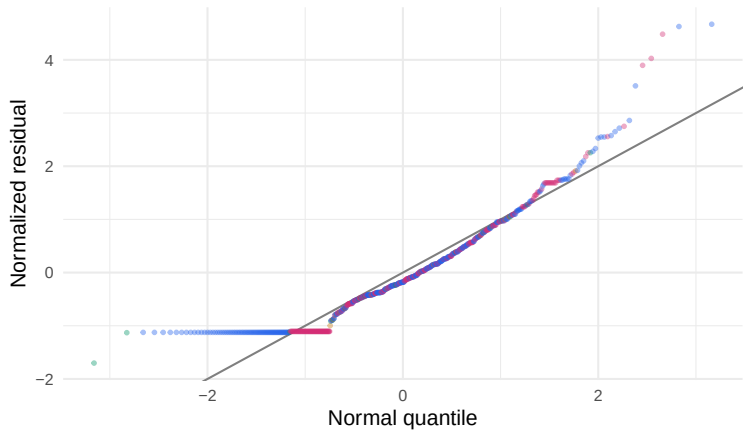
Offset 0 | MA1

Normalized residuals versus fitted



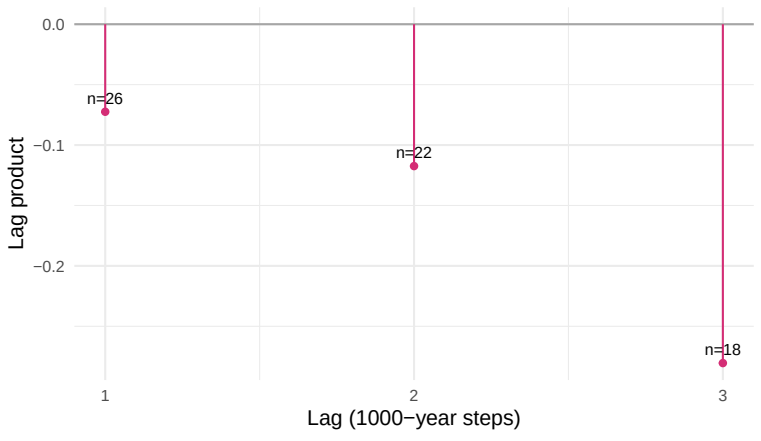
Offset 0 | MA1

Normal Q−Q; pooled normalized residuals



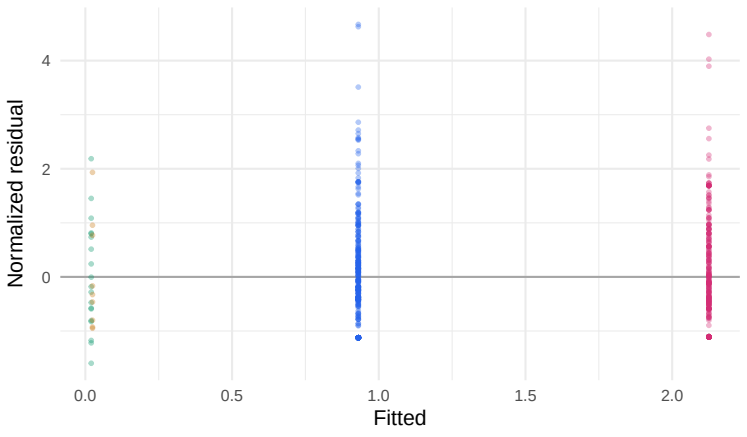
Offset 0 | MA1

Holocene residual lag products



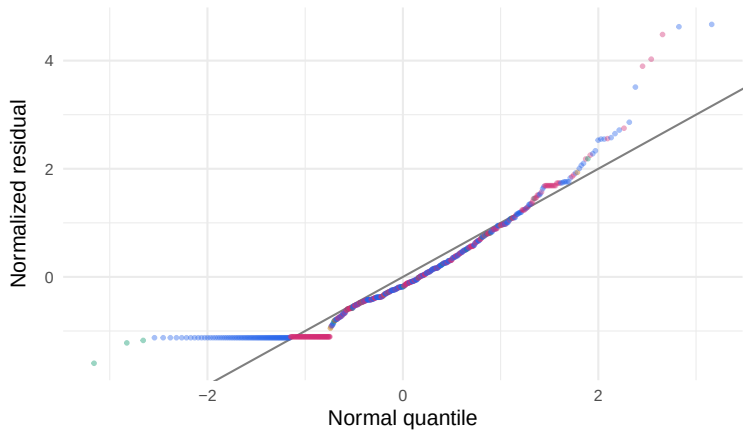
Offset 100 | MA1

Normalized residuals versus fitted



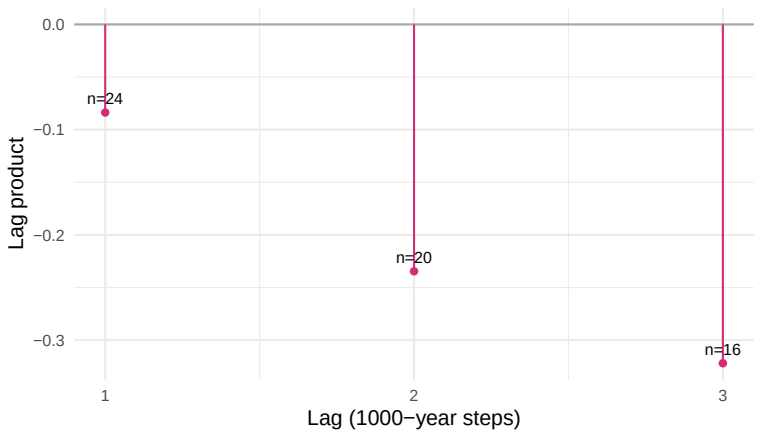
Offset 100 | MA1

Normal Q−Q; pooled normalized residuals



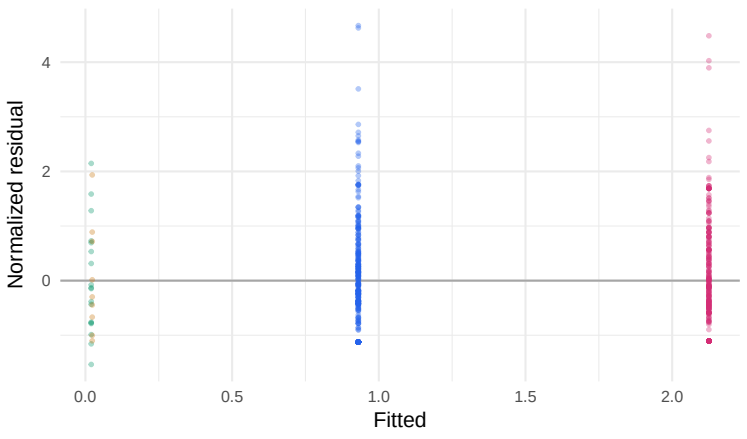
Offset 100 | MA1

Holocene residual lag products



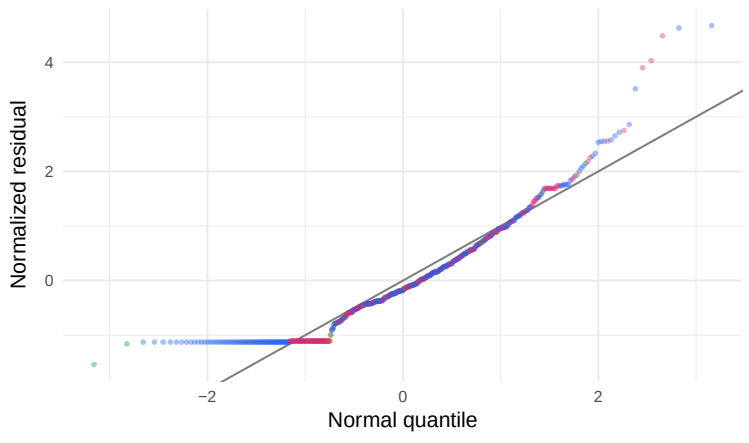
Offset 200 | MA1

Normalized residuals versus fitted



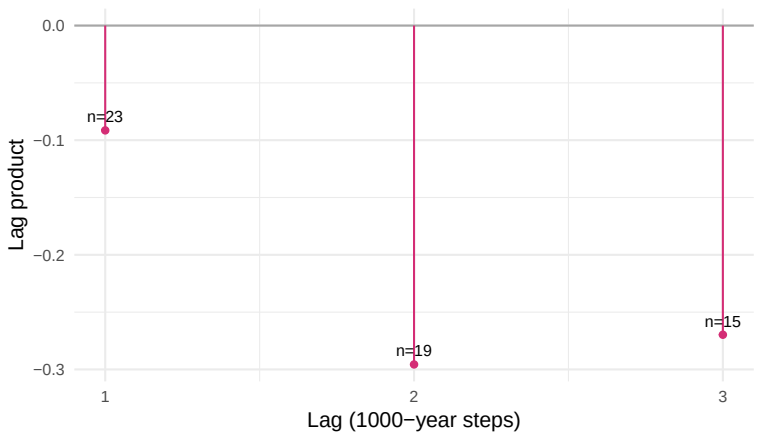
Offset 200 | MA1

Normal Q−Q; pooled normalized residuals



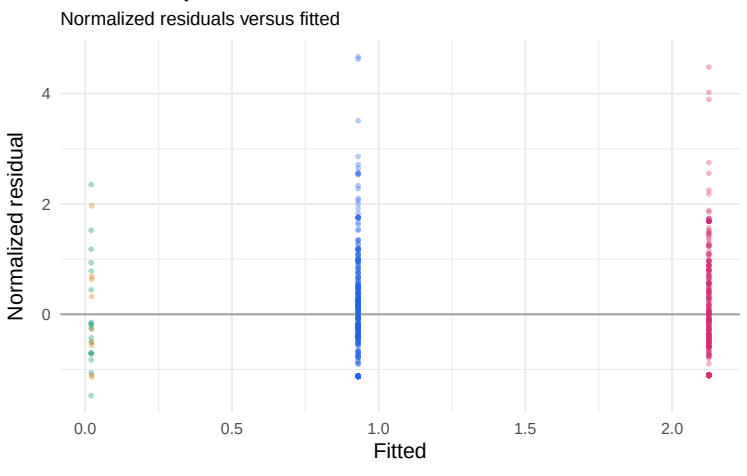
Offset 200 | MA1

Holocene residual lag products



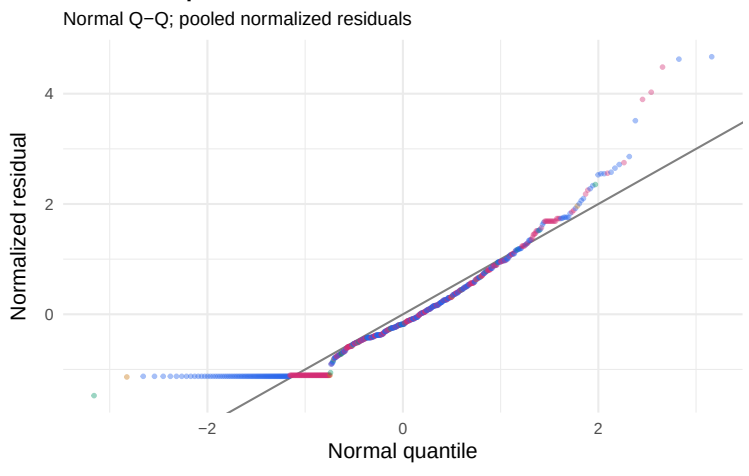
Offset 300 | MA1

Normalized residuals versus fitted



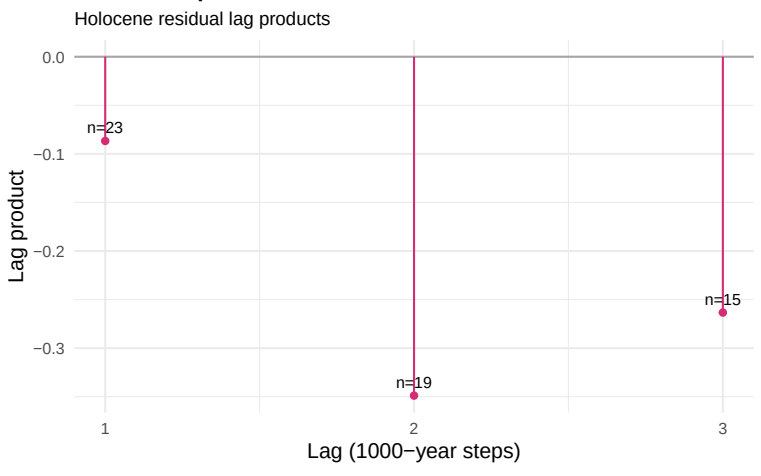
Offset 300 | MA1

Normal Q−Q; pooled normalized residuals



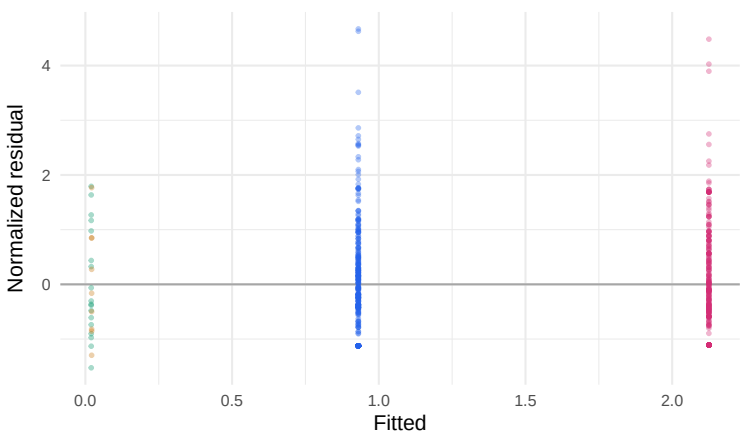
Offset 300 | MA1

Holocene residual lag products



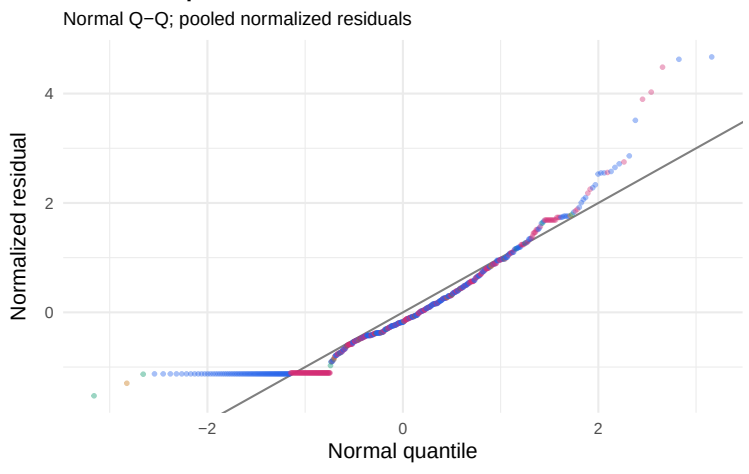
Offset 400 | MA1

Normalized residuals versus fitted



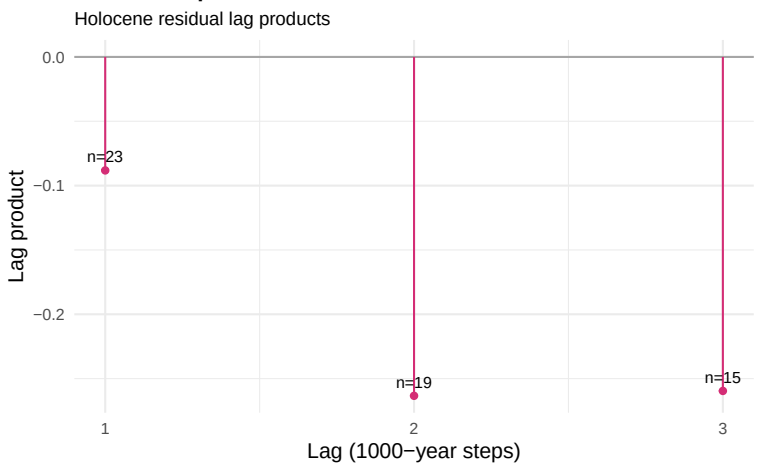
Offset 400 | MA1

Normal Q−Q; pooled normalized residuals



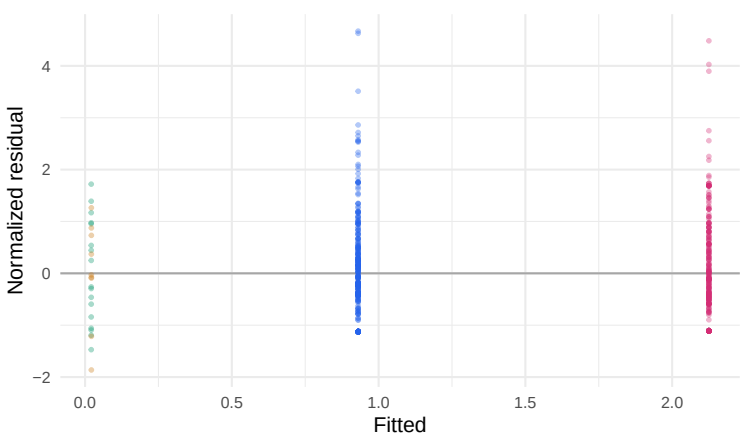
Offset 400 | MA1

Holocene residual lag products



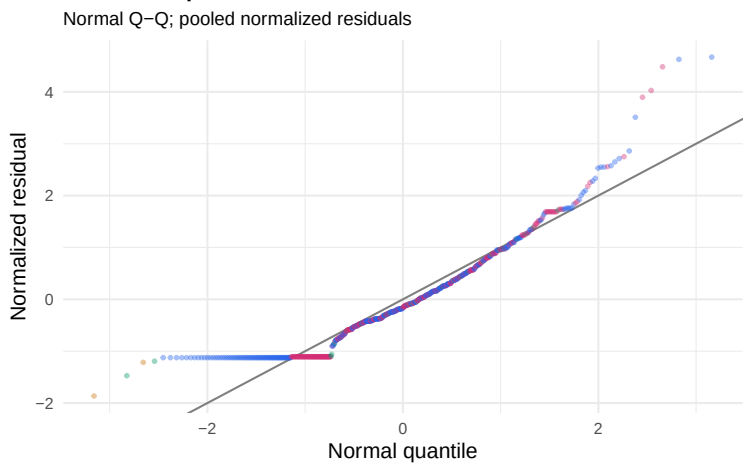
Offset 500 | MA1

Normalized residuals versus fitted



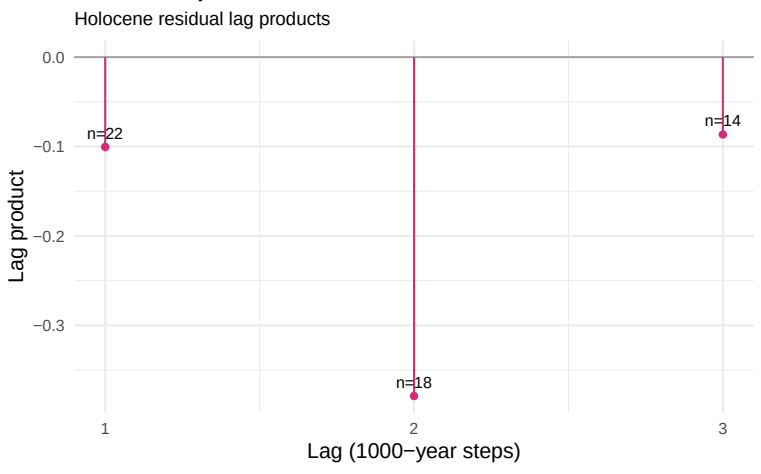
Offset 500 | MA1

Normal Q−Q; pooled normalized residuals



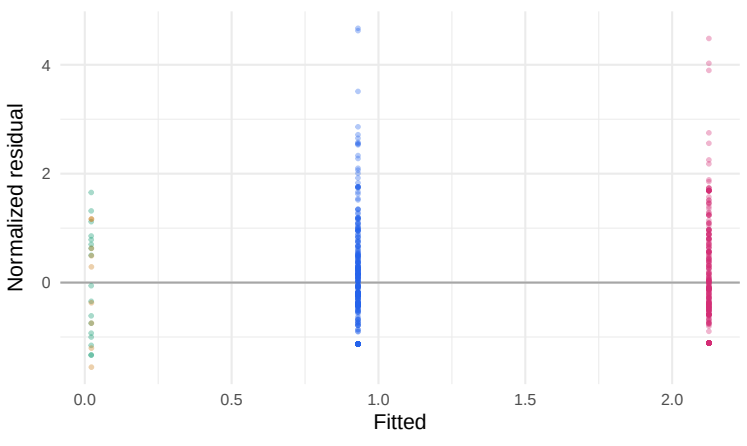
Offset 500 | MA1

Holocene residual lag products



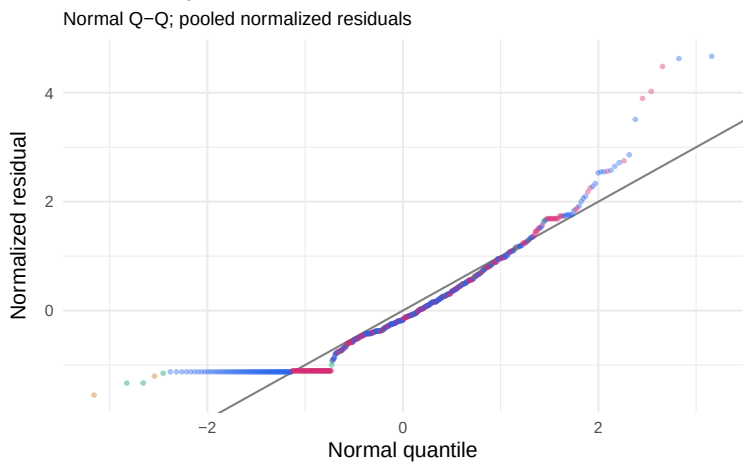
Offset 600 | MA1

Normalized residuals versus fitted



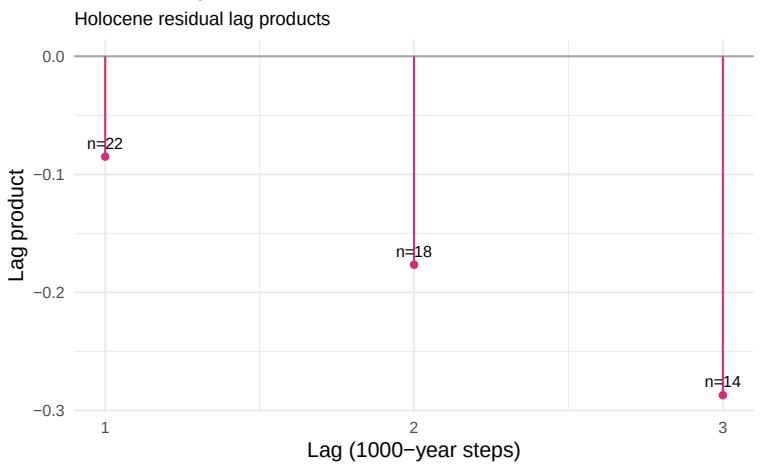
Offset 600 | MA1

Normal Q−Q; pooled normalized residuals



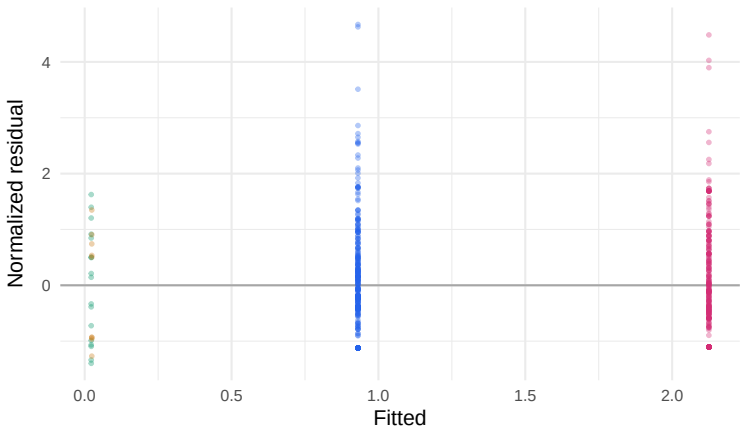
Offset 600 | MA1

Holocene residual lag products



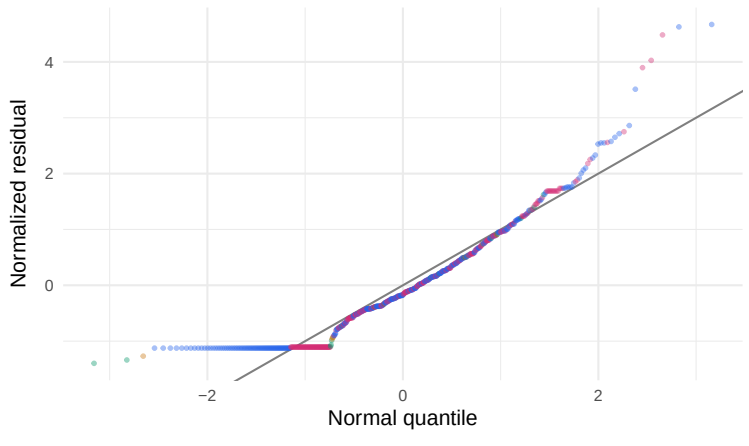
Offset 700 | MA1

Normalized residuals versus fitted



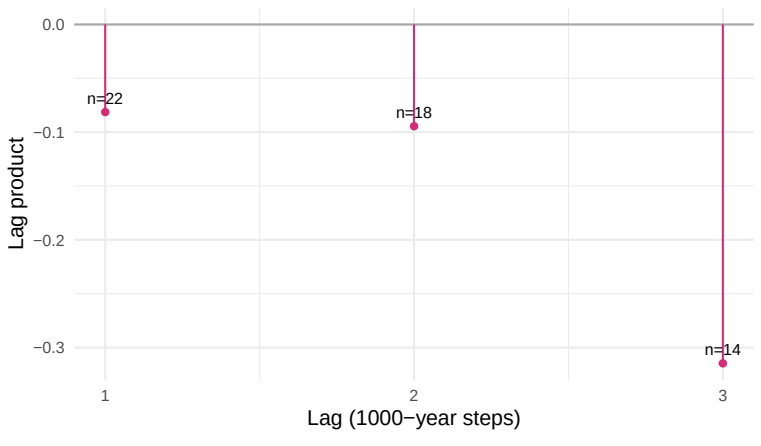
Offset 700 | MA1

Normal Q−Q; pooled normalized residuals



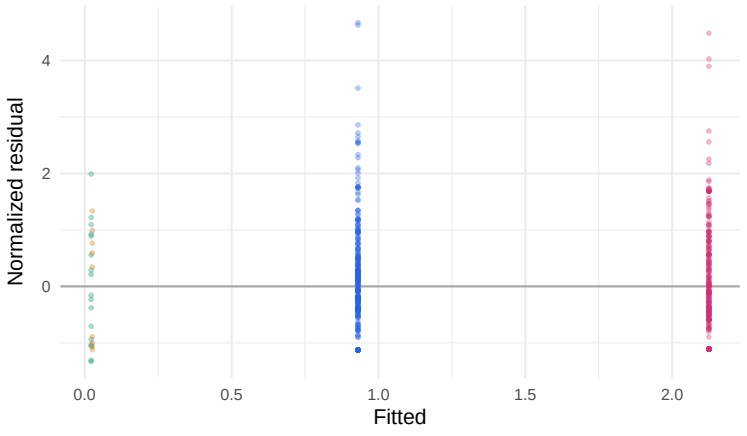
Offset 700 | MA1

Holocene residual lag products



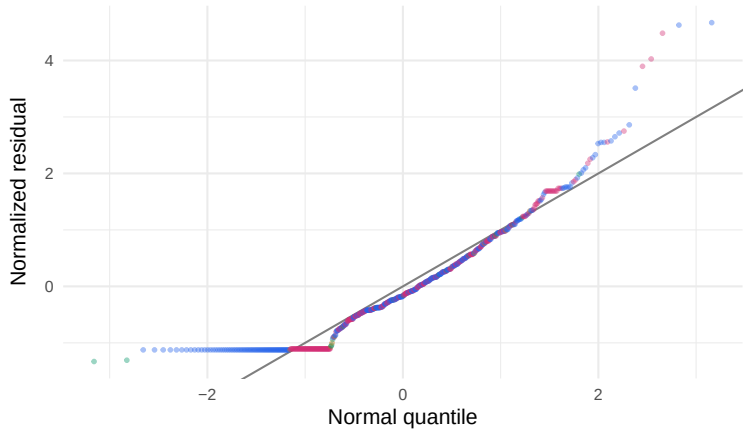
Offset 800 | MA1

Normalized residuals versus fitted



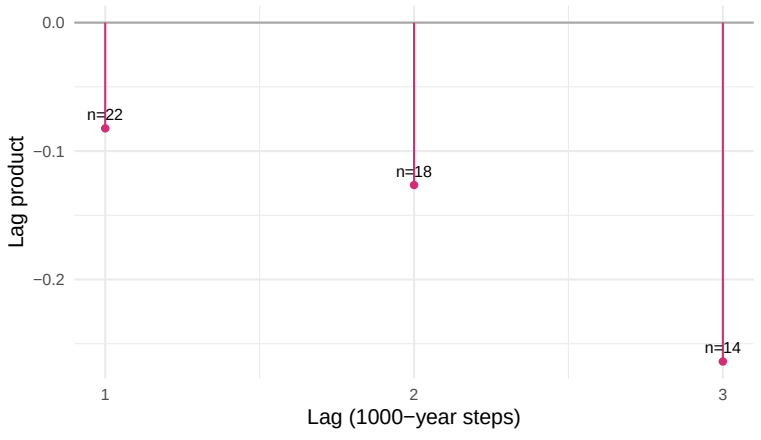
Offset 800 | MA1

Normal Q−Q; pooled normalized residuals



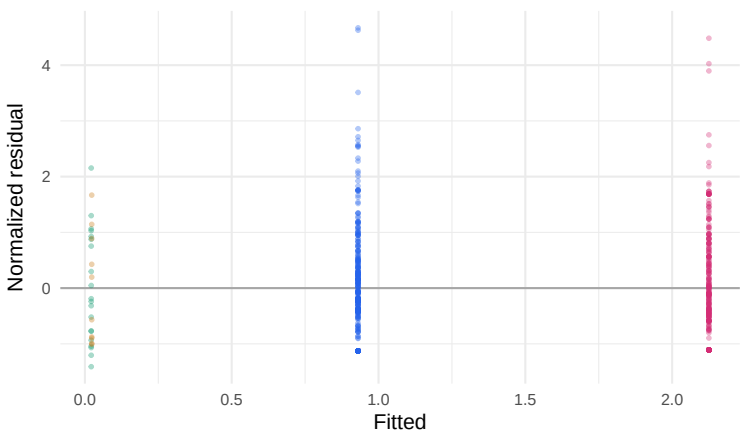
Offset 800 | MA1

Holocene residual lag products



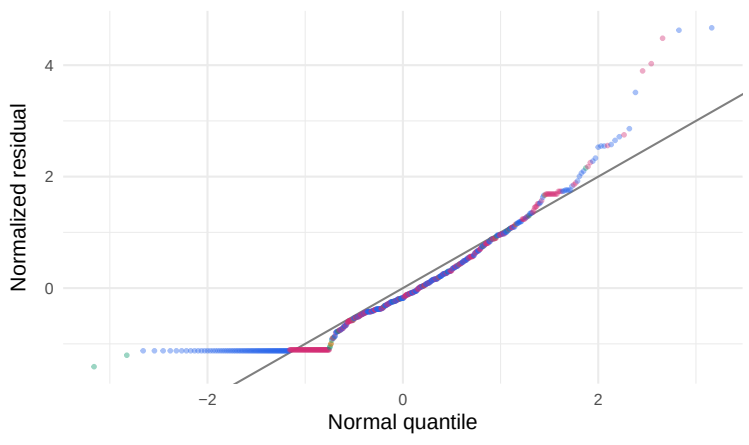
Offset 900 | MA1

Normalized residuals versus fitted



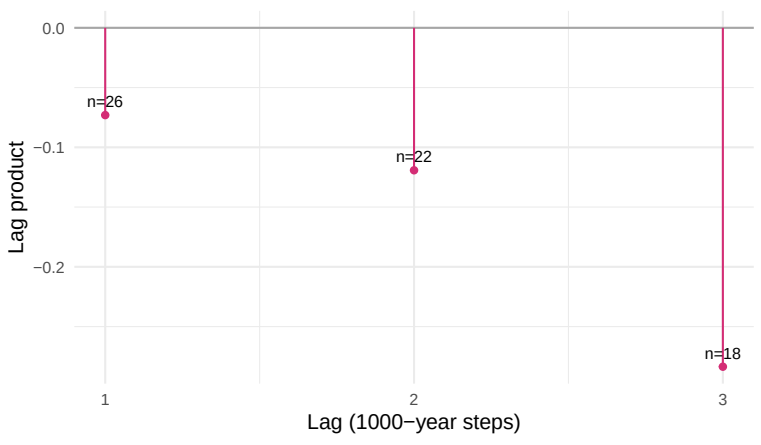
Offset 900 | MA1

Normal Q−Q; pooled normalized residuals



Offset 900 | MA1

Holocene residual lag products



Colors: Holocene MORA green, Holocene NOCA orange, Anthropocene MORA blue, Anthropocene NOCA pink.
Holocene lag products match actual within−lake time gaps; labels give pair counts. Modern singletons are excluded. No formal lag tests or confidence bands.
MA(1) selected; distribution and inference remain exploratory. Chronology uncertainty is not propagated.